Ian Todd
Sydney Medical School
University of Sydney
Sydney, NSW, Australia
itod2305@uni.sydney.edu.au

February 10, 2026

Professor A.U. Igamberdiev
Editor, BioSystems

Dear Professor Igamberdiev,

Please find enclosed the revised manuscript BIOSYS-D-25-00981, "Coherence time in biological oscillator assemblies bounds the rate of state registration," for consideration as a revised submission to *BioSystems*.

I thank both reviewers for their constructive feedback. All requests have been addressed in full.

Response to Reviewer #1

Reviewer: "The speed of thinking is restricted by the Landauer and Margolus-Levitin bounds... This should be discussed in the revised version."

Response: Section 2.3 ("Quantum and thermodynamic limits of computation") now explicitly addresses the Landauer bound, the Margolus-Levitin quantum speed limit, and the energy-time uncertainty relation, showing that coherence time exceeds quantum speed limits by eleven orders of magnitude in biological neural systems. Multi-module phase coordination—not quantum switching speed or Landauer dissipation—determines processing rate.

Response to Reviewer #2

Reviewer: "Speed of thought should be changed to speed of neural processes associated with thinking."

Response: "Speed of thought" has been replaced with "neural processes associated with cognition" throughout, including the abstract.

Reviewer: "Quantum limits of information transmission based on energy-time uncertainty relation should be discussed."

Response: Energy-time uncertainty is discussed in §2.3, following the framework developed by Igamberdiev (2021, 2025, 2026).

Reviewer: Cite four specific BioSystems papers (DOIs provided).

Response: All four are now cited: Igamberdiev (2021, 104471), Igamberdiev (2025, 105451), Igamberdiev (2026, 105707), and Khrennikov et al. (2025, 105573).

Reviewer: "Abstract should not contain references. Also, if formulas cannot be avoided in abstract, it would be better that you simplify them wherever possible."

Response: The abstract (217 words) contains no citation numbers. Mathematical notation has been simplified; only simple inline symbols (M, r, ε) are retained for precision.

Additional improvements

Beyond the reviewer requests, the revision includes: unified notation ($\lambda_{attempt}$, s^{-1}); improved Kuramoto validation methodology (Kaplan-Meier survival analysis, theory-consistent fitting, $\hat{\alpha} = 0.99$, $R^2 = 0.97$); sensitivity analyses for module-size confound and dwell-time threshold; explicit discussion of participation ratio vs. scaling exponent; new sections on frequency-dimensionality (§4.5, with spatial extent argument and four empirical predictions) and subjective temporal scaling (§4.6, with pharmacological predictions across four drug classes); two new figures; and an expanded limitations section addressing parameter identifiability and regime boundaries. The manuscript has grown from 41 to 47 pages. All simulation code is available at `https://github.com/todd866/coherence-time`.

Sincerely,

Ian Todd
Sydney Medical School
University of Sydney